Chandak Chakma

Research Assistant, Cortex AI Lab

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Research Interests

- **Self-evolving LLM reasoning:** label-free self-play in which a Challenger proposes questions and a Solver learns from them; the validity of model-generated questions; rewards that a learning model cannot game.
- **RL for LLM reasoning and verifiers:** RL with verifiable rewards (RLVR), reward design beyond ground-truth labels, LLM judges, and the pass@1 vs. pass@k trade-off.
- **Probabilistic evaluation of world models:** whether sampled futures match the real spread of outcomes, and how to train models whose futures do.

Education

B.Sc. in Robotics and Mechatronics Engineering *Sep 2022 – Nov 2026*
University of Dhaka Dhaka, Bangladesh

Research Experience

Research Assistant *Aug 2026 – Present*
Cortex AI Lab, University of Dhaka Dhaka, Bangladesh

- First-authored a NeurIPS 2026 workshop paper (accepted) on RLVR and a CVPR 2027 submission on video world models; now leading an ongoing project on self-evolving LLM reasoning.
- Co-authored three more CVPR 2027 submissions (latent world-model planning, surgical video generation).
- Built pre-registered evaluation pipelines with sealed holdouts and paired bootstrap CIs, and ran every study on a single RTX 3090 Ti (GRPO on Qwen2.5, Qwen3 Challenger–Solver loops, world-model fine-tuning).

Research Student *Apr 2025 – Present*
Robotics Lab, University of Dhaka (advisor: Dr. Sejuti Rahman) Dhaka, Bangladesh

- Undergraduate thesis: trained a maneuver-agnostic recovery policy for a rover trapped in granular regolith, using a grain-level material point method (MPM) simulator; accepted as a poster at an IROS 2026 workshop.
- Built simulator checks; an audit caught a coupling-buffer fault that silently dropped 9.7% of contact nodes.

Assistant AI Engineer (Contract) *Oct 2025 – Apr 2026*
NoeticX Remote, USA

- Built a Bangladeshi NID card OCR API (PaddleOCR, OpenCV, FastAPI) that returns name, date of birth and NID number as structured JSON.
- Developed a tablet-strip OCR pipeline that reads drug data from medicine packaging under real-world noise.
- Created a natural-language CSV data analyst (PandasAI, Plotly) for querying and plotting data in plain language.

Ongoing Research

Valid-Zero: Validity-Gated Self-Play for LLM Reasoning *2026 – Present*
Qwen3 models, Challenger–Solver self-play, LLM judges and pre-registered evaluation

- Found that about half (49%) of the questions an uncertainty reward selects in R-Zero-style self-play are invalid, and that this drift needs a learning Solver (8% invalid with a frozen Solver vs. 37% with a training one).
- Built Valid-Zero, which gates the Challenger's reward with a frozen reader of the question text; in a pre-registered loop it cut invalid questions in the Solver's data from 37% to 10% (filtering alone: 21%) at equal accuracy.
- Showed its limit: over 10 rounds the Challenger learns to pass a frozen reader (10% to 29% invalid); current work tests readers that update inside the loop.

Selected Publications

Best-of-K Can Misrank Video World Models
Chandak Chakma, Syed Nazmus Sakib, Abdul Monaf Chowdhury, Nafiul Haque, Md Mehedi Hasan, Shifat E. Arman
***Under review**, CVPR 2027*
Shows that best-of-K, the standard score for one-pass video world models, can rank the worse model first: best-of-many training raises it while proper scores worsen, and best-of-100 misranks 2 of 4 published model pairs. Proposes a matched-noise control and ALE-WI, a fine-tuning that beats widening by 2–3% in CRPS.

Unlearnable, or Unmeasured? On the Reliability of Difficulty Labels in RLVR
Chandak Chakma, Syed Nazmus Sakib, Nafiul Haque, Shifat E. Arman
***Accepted**, NeurIPS 2026 Workshop on Transitioning from Pre-Training to Post-Training* [arXiv]
Shows that "unlearnable" prompt sets in RLVR depend on measurement choices (the same data yields 74–228 prompts), while the slow-learning effect survives at about one third of the normal rate; gives a binomial model for the rollout budget a reproducible difficulty label needs.

Escape Without a Recipe: Maneuver-Agnostic Mars Rover Recovery from Granular Entrapment
Meraj Hossain Promit, **Chandak Chakma**, Md Jubair Ahmed Sourov, Sejuti Rahman
***Accepted (Poster)**, IEEE IROS 2026 Workshop (SRW), Pittsburgh* [pdf]
A recurrent policy trained in a grain-level simulator frees a trapped rover without scripted maneuvers, escaping 62.5% of verified traps on average against 7/32 for the best of ten scripted controllers.

Graph Anchors: Training-Free Long-Horizon Planning with Latent World Models
Syed Nazmus Sakib, Abdul Monaf Chowdhury, Peizhen Li, Nafiul Haque, **Chandak Chakma**, Md Mehedi Hasan, Zehong Cao, Shifat E. Arman
***Under review**, CVPR 2027*
Gives a frozen latent world-model planner a nearby target on a graph of its own training data instead of a far goal; with no training, success at 200 steps on PointMaze-Medium rises from 13% to 85%.

Contingency Planning with Latent World Models under Irreversible Outcomes
Abdul Monaf Chowdhury, Syed Nazmus Sakib, Ahanaf Hasan Ariq, **Chandak Chakma**, Nafiul Haque, Shifat E. Arman, Md Mehedi Hasan
***Under review**, CVPR 2027*
Plans with a frozen latent world model by sampling several futures per plan, scoring them with a learned asymmetric time-to-goal that makes irreversible failures costly, and branching on future observations; best average success on nine stochastic tasks.

What-If-Surg: Controllable Generation of Counterfactual Tool–Tissue Interactions in Laparoscopic Surgery with Video Diffusion
Nafiul Haque, Syed Nazmus Sakib, Abdul Monaf Chowdhury, **Chandak Chakma**, Peizhen Li, Shamim Ahmed Deowan, Md Mehedi Hasan, Zehong Cao, Shifat E. Arman
***Under review**, CVPR 2027*
A two-stage video diffusion framework that re-imagines a real surgical scene with a different instrument along a chosen trajectory, and generates the tissue response specific to that instrument (a clip, a cut or a burn).

Engineering Projects

Production RAG + ReAct Agent
Python, LangChain, FastAPI, Docker GitHub
- Hybrid retrieval (dense + BM25 with Reciprocal Rank Fusion), cross-encoder reranking, OpenTelemetry tracing and RAGAS evaluation.

Autonomous Research Agent
LangGraph, GPT-4o, Pinecone, Python GitHub
- Literature-search agent that searches the web, stores knowledge in Pinecone and writes structured reports, with human-in-the-loop approval.

LLM Hallucination Detection
Python, NLP, LLMs
- Detects hallucinations by splitting LLM outputs into claims and checking each claim against source documents.

Technical Skills

Programming	Python, C/C++
LLM Post-Training	RLVR (GRPO), Challenger–Solver self-play, LLM-as-judge, SFT, LoRA/QLoRA
LLM Apps & Agents	RAG, LangChain, LangGraph, Pinecone
ML / DL	PyTorch, Hugging Face Transformers, TensorFlow, NumPy, Matplotlib
Evaluation & Statistics	Proper scoring rules (CRPS, energy score), pre-registered tests, bootstrap CIs
World Models & Vision	Video world models, OpenCV, PaddleOCR
Engineering	FastAPI, Docker, Git/GitHub, competitive programming (3+ years)

Honors & Awards

2nd Runner-up | Datathon Bitfest, Machine Learning Competition, KUET (140 teams) 2025
28th Place | Battle of Brains Programming Contest, University of Dhaka (as a first-year) 2023